# Notes on plain futurates: Modality, description, and causation

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**Abstract.** Plain futurates—bare-verb or simple-present sentences that assert a scheduled future event—raise a version of Aristotle's puzzle of future contingents: how can a modal-free sentence make a determinate, assertable claim about an event whose truth is not yet settled? A prominent answer posits a covert future-oriented modal that does silently what *will* does overtly. This paper argues against that answer and sketches an alternative. Drawing on Mandarin and English data, we show that plain futurates systematically fail the standard diagnostics for covert modality. We then argue that a modal-free description analysis, on which a futurate merely describes a currently-holding schedule, is too weak to capture the speaker's commitment to the event's occurrence. As an alternative, we propose that a plain futurate asserts a simple proposition and presupposes that the speaker's knowledge is **causally sufficient** for it, and we motivate this direction by highlighting a connection between plain futurates and dispositional statements.

**Keywords:** Plain futurates · Covert modality · Description · dispositional statements · Causal model.

## 1 Introduction

Aristotle's puzzle of future contingents concerns whether statements about the future can bear a truth value here and now. Consider the classic example in (1).

(1) Tomorrow, there will be a sea battle. (*De Interpretatione*, Ch. IX)

If (1) is already true or false today, the openness of the future appears to be undermined; yet a speaker who *asserts* a declarative sentence with content $p$ thereby becomes committed to acting as though $p$ is true. If a future statement lacks a truth value, why is such a statement assertable at all?

A prominent line of work answers this by taking future statements to be inherently *modal* [4, 8, 3] (cf. [30, 20, 22]). Modals fundamentally evaluate a proposition with respect to non-actual worlds—a property often called *displacement*. On this view, *will* in (1) requires the proposition in its scope to hold across all possible worlds representing the normal development of the current situation. The modal analysis finds support in the observation that root modals are typically future-oriented, as in (2).

(2) a. I **will** help you. (bouletic)
b. You **may** leave after the exam. (permission)
c. He **can** do the experiment. (ability)
d. You **must** complete the assignment. (obligation)

However, not every future statement involves an overt modal. In many languages a bare verb or the simple present can assert a scheduled future event, as shown in the following examples. To save space, English data are presented as the free-translation line of the corresponding Mandarin example: throughout the paper the English translation is itself a grammatical English counterpart of the Mandarin sentence and instantiates the same phenomenon, so a single numbered example furnishes data from both languages. Where the two languages diverge, for instance, in the tense-compatibility facts of Section 3, the relevant contrast is stated explicitly and English-only examples are given separately.

(3) Húrén xià-zhōu **yíngzhàn** Yǒngshì.
Lakers next-week play Warriors
'The Lakers **play** the Warriors next week.'

Uttered at speech time on a certain day, such as Oct. 18, (3) is true iff the scheduled game occurs at the relevant future time in the next week. Its truth value is thus not determined at the speech time—it can be settled only when the game actually takes place—yet the sentence is fully assertable. Such sentences are known as **plain futurates** [4, 5, 16, 39, 13], to be distinguished from English progressive futurates (e.g., *the Lakers are playing the Warriors next week*). Plain futurates thus pose, in an especially sharp form, the question raised by the sea-battle puzzle: how can a bare, modal-free sentence make a determinate and assertable claim about a future event?

The natural first hypothesis is that the modality is present but silent: a plain futurate contains a **covert** future-oriented modal that does semantically what *will* does overtly [4, 5, 16, 39, 13]. While positing a covert operator is a common strategy for handling semantic puzzles, this paper shows that the covert-modality hypothesis for plain futurates lacks solid linguistic support. Given that the covert modal is not justified empirically, this paper examines a potential modal-free account proposed by [35], on which a plain futurate merely describes a currently-holding schedule, arguing that this description analysis is too week: a speaker who asserts a plain futurate is committed not just to the schedule being as described but to the scheduled event's *coming about*, a commitment that description alone does not supply.

Instead of developing a complete account, this paper highlights a connection between plain futurates and dispositional statements, such as *This mushroom is poisonous*, which can be asserted on the strength of a kind-level or lawlike regularity rather than direct observation. Built on a causation-based analysis of dispositional statements advocated by [33], I suggest an alternative route of understanding plain futurates:

(4) A plain futurate simply asserts a proposition $\phi$ but presuppose that the speaker's knowledge at the speech time is **causally sufficient** for $\phi$.

Accordingly, plain futurates do not involve a covert modal; they are assertable because the speaker's causal knowledge is sufficient to settle the truth of the future statement.

The paper is organized as follows. Section 2 sets out the research background, distinguishing plain futurates from other uses of the bare/present form and reviewing the covert-modal ($\mathbb{PLAN}$) analysis. Section 3 presents a dilemma: assigning a determinate truth value seems to require modality, yet plain futurates fail the standard diagnostics for covert modality. Section 4 turns to the description analysis of [35] and argues that describing a schedule does not capture a speaker's commitment to the scheduled event's occurrence. Section 5 draws the connection to dispositional statements and develops the causation-based direction. Section 6 concludes.

## 2 Background

In Mandarin, bare-verb sentences typically receive an *imperfective* reading: the event or state is ongoing and holds at the speech time, similar to the English simple present. The imperfective use also subsumes continuous, habitual, and generic readings, as illustrated in (5)–(7).

(5) Lǐbái zhù zài Shànghǎi.
Libai live at Shanghai
'Libai lives in Shanghai.' (continuous: an existing, continuing state)

(6) Lǐbái (jīngcháng) hē kāfēi.
Libai often drink coffee
'Libai (often) drinks coffee.' (habitual: an existing, continuing habit)

(7) Dàxióngmāo chī zhúzi.
panda eat bamboo
'Pandas eat bamboos.' (generic: an existing, continuing property)

Intriguingly, the very same bare-verb/present form can also convey a **scheduled future** event, as already seen in (3). It is this use—the plain futurates—that is the focus of this paper.

Plain futurates occur with a variety of lexical aspects, provided the future event or state they describe is in some way *scheduled* at the speech time [9, 14, 10, 11, 4, 5, 39, 40, 13], illustrated by the following Mandarin/English examples.

(8) Context: The speaker checked their schedule for the many events they had the next day.

Wǒ míngtiān hěn **máng**.
I tomorrow very busy
'I'm very **busy** tomorrow.' (stative verb)

(9) Context: The speaker knows that Libai has been asked to visit a business partner the following day.

Lǐbái míngtiān **qù** Běijīng.
Libai tomorrow go Beijing
'Libai **goes** to Beijing tomorrow.' (active verb)

(10) Context: The speaker checked the schedule of UA-01 and said:

UA-01 míngtiān xiàwǔ **dǐdá** Jiùjīnshān.
UA-01 tomorrow afternoon arrive SF
'UA-01 **arrives** in SF tomorrow afternoon.' (achievement verb)

(11) Context: The speaker is reporting someone's schedule for tomorrow:

Tā míngtiān 9 diǎn **chī-wán** zǎofàn, 10 diǎn **shàng-wán**
he tomorrow 9 o'clock eat-finish breakfast 10 o'clock take-finish
dìyī-mén kè.
first-CL class
'He **finishes** breakfast by 9 tomorrow and **finishes** his first class by 10.' (accomplishment verb)

Beyond scheduled events, plain futurates may also express predictions grounded in lawlike regularities, as shown in (12)–(13).

(12) Context: The Yellow River freezes over every December. Someone said in November:

Huǎnghé xià-gè yuè **jiébīng**.
yellow.river next-CLF month freeze
'The Yellow River **freezes** next month (, as it does every winter).'

(13) Context: In California, the sun doesn't set until 7 p.m. in the summer. Someone said in the morning of a summer day in California:

Jīnwǎn qīdiǎn tiān **hēi**.
tonight seven.clock sky dark
'The sun **sets** at 7 p.m. in the evening.'

Once the 'schedule' and 'lawlike' conditions are not met, a root modal such as Mandarin *huì* 'will' or English *will* becomes necessary to make a statement about the future.[1]

[1] It is worth noting that bare-verb or simple-present clauses also receive a future interpretation when embedded in the antecedents of conditionals, as in *If John gets a raise, he'll be happy.* Much prior work has treated these subordinated clauses as a special case of plain futurates [16, 37, 36]. However, subordinated clauses of this kind do not bear the same constraint as matrix plain futurates: they can express unschedulable future events without any overt future modal, as shown in below.

(i) Rúoguǒ Yǒngshì xiàzhōu jībài Húrén, tāmēn jiù nèng
if the.Warriors next.week defeat the.Lakers they just can
dǎ-jìn zǒngjuésài.
fight-enter final

(14) Yǒngshì xiàzhōu #(huì) jībài Húrén.
the.Warriors next.week will defeat the.Lakers
'The Warriors #(will) defeat the Lakers next week.'

(15) Tā míngtiān #(huì) hěn kāixīn.
he tomorrow will very happy
'He #(will) be happy tomorrow.'

(16) Tā míngzǎo #(huì) chídào.
he tomorrow.morning will late
'He #(will) be late tomorrow morning.'

A well-motivated response to the puzzle is to take plain futurates to contain a covert modal, $\mathbb{PLAN}$, which allows the relevant proposition to be evaluated across possible developments of the actual world $w_\circ$ [4, 5, 16, 13] and the speech time $t_\circ$. On this view, the meaning of (3) can be modeled as follows.

(17) a. Presupposition
$[\![\mathbb{PLAN}$ [the Lakers play the Warriors next week]$]\!]^{w_\circ,t_\circ}$
**is defined only if** the Lakers–Warriors game is scheduled for next week, i.e., at some time after $t_\circ$, in $w_\circ$;
b. At-issue content: if defined, ...
$[\![\mathbb{PLAN}$ [the Lakers play the Warriors next week]$]\!]^{w_\circ,t_\circ}$
**is true iff** for all $w$ accessible from $w_\circ$, the scheduled event is not interrupted in $w$, there is a time $t$ such that $t_\circ < t$, $t$ is included in the next week, and the Lakers play the Warriors at $t$ in $w$.

The covert modal thus efficiently captures the meaning of plain futurates and aligns them with other future-oriented statements. Section 3 argues that, despite this appeal, the covert-modal analysis is empirically problematic.

## 3 Lack of linguistic evidence for the covert modal

To assign a determinate truth value to a plain futurate, incorporating modality appears necessary. If plain futurates really involved a covert modal, they should at least display the same linguistic patterns as typical modal constructions. However, they do not. This section presents three representative diagnostics—modal *wh*-indefinites, modal subordination, and futuer in the past.

**Modal sentences as a licensing environment.** In natural languages, the distribution of certain lexical items is contingent on the presence of a modal:

'If the Warriors beat the Lakers next week, they will make it to the Finals.'

Mendes [23] argues that the future found in these subordinate environments is inherently different from the scheduled future found in matrix clauses. The evidence comes from Portuguese, which has a morphologically distinct subjunctive future form in conditional antecedents, whereas scheduled and habitual readings use the ordinary indicative present. Accordingly, this paper sets these subordinate future clauses aside.

such items are licensed only when a modal is available in their local environment. This dependency furnishes a diagnostic for covert modality. If plain futurates contained a covert future modal, they should be able to license these modal-sensitive items just as overt modals do. A case in point is the Mandarin *wh*-expression, which may be construed either interrogatively or existentially. On its non-interrogative, existential construal it typically requires a licensor, prototypically a modal. In (18), for instance, the *wh*-expression is interpreted as somewhere rather than as a question word. This licensed use is known as the 'modal *wh*-indefinite'.

(18) Lǐbái míngnián {kěyǐ/ yào/ huì} qù gè **shénme** dìfāng dùjià.
Libai next.year can/ want/ will go CL what place vacation
'Libai can/wants to/will go somewhere for vacation next year.'

However, plain futurates cannot support this use, as shown below. (19) can only be interpreted as a question.

(19) # Lǐbái míngnián qù gè **shénme** dìfāng dùjià.
Libai next.year go CL what place vacation
Intended: 'Libai goes somewhere for vacation next year.'

One might object that covert modals differ from overt ones precisely in being unable to serve as licensors. This objection is hard to sustain. Using corpus data, Liu and Yang [21] argue that modal *wh*-indefinites can in fact be licensed by a covert epistemic modal in non-futurate sentences—specifically, ones containing overt aspectual marking, as in (20). Covert modals are therefore capable of licensing modal *wh*-indefinites. If futurates likewise contained a covert modal, it would be mysterious why they cannot.

(20) Context: B and A are talking about someone in B's church who got a doctor's license despite being an immigrant.

Éi, wǒmēn jiàohuì yǒu yí-gè rén kàn-guò shěnme
ei our church have one-CLF person take.exam-PFV what
zhízhào ... (zhīhòu) cái qù zhǎo yīshēng.
license afterwards just go be doctor
'This reminds me,our church has someone who took an exam to get some license, (andthen) went on to be a doctor.' CallFriend Mainland Mandarin Corpus, File 4227, line 494-502 ([21]: 585)

Moreover, the same holds in English: the free-choice use of *any* must be licensed by a modal, and a 'schedule' context does not rescue the modal-free variant, as (21) shows.

(21) a. According to next season's fixture list, the Lakers meet every club home and away; so **any** team #(will) play(s) the Lakers next season.
b. The invigilation rota is fixed: **any** teacher in our school #(will/can) proctor the make-up exam tomorrow.

c. According to the itinerary, we #(will/can) eat at **any** of the restaurants on the pier this afternoon.

In each case the schedule makes the future futurate independently well-formed, yet the presence of free-choice *any* forces an overt modal all the same. A covert-modal analysis of plain futurates has no explanation for this contrast.

As with modal *wh*-indefinites, free-choice *any* is also licensed in imperatives, which have been argued to involve either a covert deontic modal or at least a modal component in the imperative speech act [12, 29, 15].

(22) Pick **any** one of the five cards!

**Modal subordination.** Following [32, 17, 3], modal subordination is a hallmark of genuine modal sentences. In (23), for instance, the second *will*-sentence is interpreted as if it were subordinated to the antecedent of the preceding conditional—that is, it is evaluated relative to the worlds introduced by that antecedent. Dropping *will* makes the discourse move incoherent, because the meeting is no longer construed as contingent on our going to NYC.

(23) If we go to NYC, we will stay at the Trump Tower. The president #(will) meet(s) us there.

The same pattern holds in Mandarin: an overt modal is required to support modal subordination, as in (24). Without an overt modal, B's utterance cannot be interpreted relative to the counterfactual situation established by A.

(24) A: Yàoshì fēijī méi wǎndiǎn, wǒmēn zǎoshàng jiù dào-le.
if flight not delay we morning just arrive-PFV
'If the flight hadn't been delayed, we would have arrived in the morning.".'

B: Zhào jìhuà, dāihuì hái #(néng) chī huǒguō ne.
according.to plan later also can eat hot.pot SFP
'According to the original plan, we #(can) even eat hot pot later.'

Another version of the diagnostic involves anaphora to a discourse referent introduced under a modal, as in (25)–(26) [32] In each case the first sentence introduces a discourse referent $x$ for a person within the scope of a modal; the second sentence can retrieve $x$, and evaluate its content relative to the worlds introduced by the first sentence, only if it too contains an overt modal.

(25) Míngtiān 10 diǎn yīnggāi yǒu rén$^x$ lái. Gēnjù liúchéng,
tomorrow 10 o'clock should have person come according.to process
tā$_x$ #(huì) qīngdiǎn yí-biàn huòjià shàng de shāngpǐn.
she will check one-time shelf on MOD item
'There is supposed to be someone coming at 10 a.m. tomorrow. According to the process, they #(will) check the items on the shelf.'

(26) A: Jīnián kěnéng yǒu rén$^x$ méi guò káohé.
this.year seem have person not pass appraisal

'There seems to be a person who didn't pass the appraisal.'

B: Nà àn gōngsī guīdìng, tā$_x$ míngnián #(yào)
then according.to company regulation s/he next.year must
lízhí.
resign
'Then, according to the company regulations, they #(must) resign next year.'

The challenge for a covert-modal analysis is thus clear: if plain futurates contained a covert modal, why would they fail to support modal subordination—a property that overt modals reliably exhibit?

**Future in the past.** In English, future modals are compatible with past-tense morphology [4, 5, 35]. In (27), for instance, the past progressive futurate describes a past plan-state concerning a still-future event, the futurity being marked by *tomorrow*. Note, further, that the progressive has itself been widely argued to be modal [7, 19, 1, 28]. Thus (27)-(i) reports that, as of some past reference time, the schedule had Lowe pitching tomorrow. The plan held in the past; whether it still holds is left open, and often the sentence implicates that it has since changed. Plain futurates, by contrast, are incompatible with the past tense, as shown by (27)-(ii) shows. Embedding nonetheless rescues them, as in (27)-(iii).

(27) A: When does Lowe start next?
B: (i) Lowe was starting tomorrow against the Yankees.
(ii) #Lowe started tomorrow against the Yankees.
(iii) Jenny said that Lowe started tomorrow against the Yankees.
([5]: 39)

In [35], it is reported that specifying an old schedule can significantly improve the acceptability of futurates with the past tense, as shown below.

(28) Context: A is reading the sports news and sees that an upcoming game has been canceled due to storms. They turn to B and say:
The Major League schedule has been revised. **In the original schedule**, the Red Sox **played** the Yankees tomorrow, but now they won't.
([35]: 205)

Crucially, both (27)-(iii) and (28) are embedded under a linguistic intensional environment. Their truth values are evaluated in a world departing from the actual one: in the former, the plain futurate clause is evaluated relative to Jenny's doxastic worlds, while in the latter, the futurate clause is evaluated in the world where the schedule had not been revised.

Mandarin patterns alike: a future-oriented modal can express a future event from a past perspective, whereas a futurate cannot unless it is embedded. In (29) the past-perspective report requires *huì* 'will'; the bare futurates are acceptable when embedded by an attitude verb, as in (30), or when saliently specifying a previous schedule, as in (31).

(29) Nián qián, gōngsī hái #(huì) zài xià-gè yuè gěi wǒmen fā
year before company still will at next-CL month to us give
jiǎngjīn.
bonus
'Before the Spring Festival, the company (said they would) give us a bonus the next month.'

(30) Wǒ zhīqián yǐwéi Lǐbái míngtiān qù Běijīng. Dàn tā shì qù
I before think Libai tomorrow go Beijing but he FOC go
Shànghǎi.
Shanghai
'I thought Libai went to Beijing tomorrow, but he goes to Shanghai.'

(31) Bǐsài shíjiān gǎi-le. Àn zhīqián de rìchéng, Húrén
game time change SFP baseed.on previous schedule the.Lakers
míngtiān dǎ Huǒjiàn. Xiànzài, gǎi-dào
tomorrow fight the.Rockets now change-become
hòutiān le.
the.day.after.tomorrow
'The game schedule got changed. Based on the old schedule, the Lakers were supposed to play the Rockets tomorrow, but now it's been moved to the day after tomorrow.'

This asymmetry poses a problem for the covert-modal analysis: if plain futurates contained a covert modal, why should they fail to express the future-in-the-past that overt modals express? The point is sharpened by the embedding facts. Plain futurates can express a future-in-the-past once embedded in an intensional environment; yet if they already contained a covert modal—itself an intensional operator—it is puzzling why that modal cannot license the future-in-the-past on its own, without the help of an external embedding operator.

The covert-modal approach might, of course, be defended by holding that the modal in plain futurates is a covert modal that, unlike any overt modal, happens to license none of the usual diagnostics. But this move comes at a cost: it rescues the analysis only by stipulating away every property that would have provided evidence for it, leaving the covert modal detectable by no test whatsoever. An operator posited on these terms does no explanatory work; it merely re-describes the puzzle it was meant to solve.

## 4 Are futurates just descriptions?

Beyond the widely accepted covert-modal approach, Rullman et al [35] develop a modal-free, description-based analysis. Their central claim is that a plain futurate presupposes the existence of a schedule. A schedule is defined as a non-trivial matrix of information about predetermined future events—formally, a (physical or mental) representation that provides a set of answers to a multiple *wh*-question, the schedule question (e.g. *Who does which chore on which day?*).

A plain futurate declarative $S$ is felicitous only if the context supplies a unique, salient schedule that entials the proposition expressed by $S$. On this view, a plain futurate is not a prediction about the future but a description of the schedule that is true at the present: the present tense is licensed because the reference time is the holding time of the currently salient schedule.

The description analysis characterizes a plain futurate as “not a prediction about the future” but “a description of the schedule that is true at the present” [35]. This characterization, we argue, is too weak: a speaker who asserts a plain futurate is committed not merely to the schedule being as described, but to the scheduled event’s actually coming about.

**Commitment to the occurrence of scheduled events.** Simply describing a representation does not commit the describer to its content being true. This is clearest with quotation: in asserting (32), the speaker reports the president’s utterance without taking on any commitment to the truth of the content.

(32) The president said ‘I saved the world.’

If a plain futurate were, as the term suggests, a *description* of a schedule, we would expect the same transparency: it should be possible to describe a schedule the speaker knows to be defunct, just as one can quote a claim one believes false. But futurates do not permit this. A speaker who knows the current schedule has been revised cannot felicitously assert the plain futurate; they must involve overt modal elements, as demonstrated in (33) (see also [4, 5]).

(33) Context: A is reading the sports news and sees that an upcoming game between the Lakers and the Warriors has been canceled because the Warriors’ players are on strike for higher pay. They turn to B and say:

Húrén míngtiān #(běnlái yào) yíngzhàn Yǒngshì. Xiànzài,
the.Lakers tomorrow originally will fight the.Warriors now

bǐsài qǔxiāo le.
game cancel SFP

‘The Warriors were supposed to face the Lakers tomorrow, but now the game is canceled.’

The unacceptability of the plain futurate in (33) may be explained to be a presupposition failure—the schedule should be presupposed but is cancled in the context. However, when a schedule is intact at speech time but the speaker believes that the scheduled event will not take place, asserting a plain futurate is not felicitous, as shown in (34).

(34) Context: Dufu knows that Libai is in the process of resigning and no longer comes to work at the company. But the company is still trying to convince him to stay, and so has not yet canceled his position or his work assignments.

W: Lǐbái míngtiān qù Běijīng ma?
Libai tomorrow go Beijing QP

'Does Libai go to Beijing tomorrow?'

D: #Tā míngtiān qù. Dàn tā yǐjīng tíchū lízhí le. Bù
he tomorrow go but he already submit resignation SFP not
zhīdào gōngsī néng-bù-néng liú-zhù tā.
know company can-not-can stay-live him
'He does. However, he already submitted a resignation. It's not clear if the company can convince him to stay.'

Rullmann et al [35] are not unaware of this commitment; they build it in as a *correctness presumption*—"in the absence of explicit reference to a schedule, the default schedule … is presumed to be correct—that is, the events it describes will actually take place" ([35]: 205). This is an admission that, once asserting a plain futurate, the speaker is committed to its truth. Hence, the correctness presumption is thus an extra stipulation bolted onto the description account to recover the very commitment that "description" was meant to avoid.

Once this correctness presumption is granted, a plain futurate no longer merely describes a schedule; it describes a schedule whose events, from the speaker's perspective, will actually take place. But this revives the very problem of future contingency raised in Section 1: how can a speaker be committed to the truth of a future event without a covert modal to underwrite that commitment? The description analysis, having built the speaker's commitment into the correctness presumption, inherits the puzzle it was meant to dissolve.

**Planned outcomes still require modals.** The previous cases involved a schedule that was revised or externally affected. A yet more telling case keeps the plan fully intact and salient, yet still blocks the plain futurates. Consider a World Cup group stage. Spain's coaching staff have formulated an explicit strategy for the match two days hence: beat Portugal, draw France, and thereby guarantee advancing from the group. This strategy is a decided, common-ground plan about future events, which is precisely the kind of "information about predetermined future events" in the sense of [35]. Now compare two ways of putting the plan into words. Describing the *fixture* needs no modal, as in (35); describing the plan's *intended results* requires the modal *yào* 'must/be.to' and is infelicitous without it, as in (36).

(35) Xībānyá duì zài zhīhòu de xiǎozǔ sài (zhōng) yíngzhàn Pútáoyá
Spain team at later DE group match in meet Portugal
duì hé Fǎguó duì.
team and France team
'In the upcoming group stage, Spain plays Portugal and France.'

(36) Xībānyá duì zài zhīhòu de xiǎozǔ sài #(yào) jībài Pútáoyá
Spain team at later DE group match must defeat Portugal
duì, tīpíng Fǎguó duì.
team draw France team
'In the upcoming group stage, Spain #(has to) beat Portugal and draw France.'

Both the fixture and the intended results are entailed by the same salient, decided strategy. If a plain futurate is licensed whenever the context supplies a schedule that entails the proposition expressed, then (36) ought to be as acceptable as (35)—the strategy entails both that Spain plays Portugal and that Spain is to beat Portugal. But only the fixture licenses a bare futurate. The schedule-description condition does not distinguish the two cases, and so over-generates.

It will not do to reply that beating Portugal is a mere outcome of a scheduled event rather than part of the schedule, on the model of the description-vs-outcome contrast that [35] draw for *I am busy tomorrow* versus *#I am tired tomorrow*. Being tired is an unplanned downstream side effect that no schedule registers. Beating Portugal is the opposite: it is the explicit, deliberated *content* of Spain's strategy. A plan can be about a result without that result becoming statable as a plain futurate. So the licensing line does not fall between "in the schedule" and "outcome of the scheduled": both the fixture and the targeted result are equally decided by agents and equally part of the salient plan.

**No contentful schedule.** The present case makes the converse point: a plain futurate can succeed while there is no contentful schedule to describe at all. If so, the futurate cannot in general be a description of a schedule, because in these cases there is nothing scheduled to describe. Consider the following example.

(37) Context: Dufu has looked at Libai's calendar for tomorrow and found it empty. It means that Libai has no appointments at the office. Zhangsan also knows that, as a personal rule, Libai does not come in on days when he has nothing scheduled.

W: Wǒ míngtiān xiǎng gēn Lǐbǎi zài gōngsī kāi-gè huì.
I tomorrow want with Libai at company open-CLF meeting
'I want to have a meeting with Libai at the office tomorrow.'

D: O, tā míngtiān bù lái gōngsī.
oh he tomorrow not come company
'Oh, he is not coming to the office tomorrow.'

The description analysis requires that the context provide a salient schedule—a nontrivial matrix of predetermined future events—that entails the proposition expressed by the futurate. Recall that Rullmann et al bar trivial, single-cell matrices from counting as schedules; a plain futurate is licensed only by a genuinely contentful one. However, in (37), the relevant fact is precisely that Libai's schedule for tomorrow is *empty*: there is no scheduled event, and hence no schedule content, that entails he is not coming with Libai's personal rule—an additional premise. Strictly speaking, the presupposition requiring the existence of a schedule fails in this case. Yet Dufu's plain futurate is fully felicitous.

Even setting aside the emptiness of the calendar, the proposition Dufu asserts, i.e., *Libai will not come in*, is not information contained in any schedule. It is an inference that combines two premises: the observed fact that tomorrow is unscheduled, and Dufu's knowledge of Libai's disposition (he stays away on unscheduled days). Neither premise is itself the content of a schedule; the disposition is a generalization about Lisi's behavior, and the empty calendar is the non-existence of a schedule. The futurate reports the conclusion of this inference, not the description of a matrix.

Rullmann et al insist that a plain futurate can state a 'higher-level description' of a schedule. For example, if the speaker finds their schedule for tomorrow is full of events, they may utter *I am busy tomorrow*. This is an inference derivable from mere description of the busy schedule. Accordingly, the counterpart of this plain futurate under the context in (37) should be (38).

(38) Lǐbái míngtiān méi shì.
Libai tomorrow not event
'Libai is free tomorrow.'

However, the plain futurate in (37) requires a further premise—Libai's personal rule—beyond the empty schedule. Therefore, its acceptability is not covered by the description analysis.

## 5 Plain futurates and dispositional statements

The preceding sections argued that neither a covert modal nor a schedule description completely captures the linguistic properties of plain futurates. We have not, however, offered a fully worked-out semantics for that condition. This is a task for future work. What this section aims to do is more modest and more useful at this stage: to identify a novel connection of plain futurates to another kind of statements—dispositional statements. Such a connection may shed light on the meaning of plain futurates.

The observation that motivates the connection is epistemological. A dispositional statement, such as the ones in (39), can be asserted without any direct evidence for the manifestation it concerns.

(39) a. This drop of liquid is flammable.
b. This mushroom is poisonous.
c. This cube is water soluble.
d. This drop of liquid is poisonous.

Specifically, the speaker may know that a specific mushroom is poisonous without eating it. Therefore, whether an object has a dispositional property may not be verified by simple observation. Following Quine's suggestion [31] that what qualifies a thing as poisonous though it is never eaten is that it is of the same kind as the things that are poisonous, one can hold with van Rooij and Schulz [34] that the truth of a disposition statement "$x$ has disposition $P$" is causally guaranteed by the facts that (i) $x$ belongs to a natural kind $k$, and (ii) $P$ is an intrinsic property of $k$. Thus *this mushroom is poisonous* is guaranteed because it is a death cap and the death cap is poisonous.

Plain futurates based on natural laws have the same profile. A speaker may felicitously assert (40) without having observed the state of the Yellow River in December, which is a state that has not yet occurred.

(40) Context: The Yellow River freezes over every December. Someone said in November:

Huǎnghé xià-gè yuè **jiébīng**.
yellow.river next-CLF month freeze
'The Yellow River **freezes** next month (, as it does every winter).'

What licenses (40) is not evidence about the particular future event but knowledge of a standing regularity, i.e., the natural law governing the freezing of the Yellow River, from which the particular case follows. This is the same epistemic route by which a disposition statement is licensed: an inference from a lawlike or kind-level regularity to a claim about a particular, as-yet-unobserved case.

That plain futurates are connected to dispositions is not a new suggestion. It is the central claim of [6], who analyzes a futurate as making reference to a dispositional structure in which a disposer holds a state that causes an eventuality of the relevant description. Her key move is to generalize the intending agent of plain futurates to a disposer that may be inanimate, a physical tendency rather than an intention, so that plain futurates like (40) become "no longer truly exceptions; they are cases of dispositional causation just like the intentional futurates" [6].

Given the parallel between plain futurates and dispositional statements, I suggest carrying the spirit of [33]'s analysis of dispositions over to plain futurates. The aim here is not a complete analysis but a direction; the formal details are left for future work. The guiding idea is simple: what guarantees the truth of a plain futurate is not a covert modal but a **causal structure**, which is established based on a schedule, a natural law, or a habit, in virtue of which the speaker's knowledge is **causally sufficient** for the outcome. By this, I mean, informally, that what the speaker knows, together with the assumption that nothing unforeseen intervenes, already settles that the event will occur: the outcome follows from the speaker's information, and no appeal to a modal is needed to bridge the gap.

Speakers know a great deal about how things in the world hang together: which conditions bring about which outcomes. Following work on causal models in semantics [27, 38, 26, 24, 25, 2], we can picture this knowledge as a web of causal relations among propositions. Take the schedule case. Ordinarily, if a game is on the schedule and nothing out of the ordinary intervenes, the game takes place: the schedule, together with the normal course of events, is enough to bring the event about. We can state this as an informal rule. Let $S$ = "the game against the Warriors is scheduled for next week," $N$ = "the world runs its normal course" (no strike, no pandemic, . . . ), and $P$ = "the Lakers play the Warriors next week."

(41) Whenever the world runs normally ($N = 1$), if $S = 1$, then $P = 1$.

A speaker who asserts the futurate (3) knows that the game is scheduled ($S = 1$) and takes the world to be running its normal course ($N = 1$). Given those two things, the speaker's knowledge is **causally sufficient** for the game: $P$ follows from what the speaker knows, with no modal needed to mediate. The same picture extends to the other cases—a natural law is what makes the speaker's knowledge causally sufficient for *The sun rises at 7AM tomorrow*, and a standing habit does so for a habitual futurate. In each case the futurate is licensed under a single condition: the speaker's knowledge, together with the assumption that nothing unforeseen intervenes, is causally sufficient for the outcome. This is the condition I propose lies at the heart of futurate meaning, and it is the same notion that grounds the dispositional statements discussed above—which is what makes the bridge between the two worth pursuing.[2]

Accordingly, the truth condition of a plain futurate can be represented as follows. A futurate sentence denotes a simple proposition $\phi$ as its at-issue content, while presupposing that the speaker's knowledge at the speech time $t_\circ$ is causally sufficient for the truth of $\phi$.

(42) $[\![\text{The Lakers play the Warriors next week}]\!]^{\mathcal{M},t_\circ,w_\circ} =$

$$\mathsf{play}(\mathsf{l},\mathsf{w},t_{\text{TOP}},w_\circ) \wedge t_{\text{TOP}} \gtrsim t_\circ \wedge t_{\text{TOP}} \subseteq \mathsf{next\text{-}week} \qquad (= \phi)$$

$$\textbf{defined only if } \mathcal{K}_{\mathsf{s}}(w_\circ,t_\circ) \models_{\mathcal{M}} \phi$$

Here $\mathcal{K}_{\mathsf{s}}(w_\circ,t_\circ)$ consists of the propositions in the speaker's knowledge at $w_\circ$ and $t_\circ$ together with their truth values relative to $w_\circ$ and $t_\circ$, and $\models_{\mathcal{M}}$ is causal entailment relative to the causal model $\mathcal{M}$. Specifically, in this case **the speaker knows both that the game is scheduled for next week and also knows a general schedule rule—a schedule, under normal conditions, brings about the event it schedules**. Note that the present tense is interpreted as marking **non-past** time ($t_{\text{TOP}} \gtrsim t_\circ$), rather than requiring strict overlap with the speech time.

Because this analysis posits no covert modal, it is unsurprising that plain futurates do not exhibit the linguistic hallmarks of modal sentences. This is not to deny that futurates are interpreted relative to the speaker's knowledge of normality conditions and causal relations—knowledge that plays a role not unlike a modal base. But this concession does not reintroduce the covert modal by the back door: the epistemic component is contributed **presuppositionally**, as a condition on when the futurate is defined, rather than as at-issue content asserted by a modal operator.[3]

This is as it should be, for the speaker's epistemic state arguably underwrites utterances quite generally, not plain futurates in particular. Kaufmann makes a related observation, though he draws a stronger conclusion: "all sentences in the bare tenses contain a covert epistemic necessity operator and are evaluated against, or predicated of, metaphysical or doxastic modal bases. Thus their truth conditions involve either settledness or belief, not merely truth simpliciter" [16, p. 246]. Where Kaufmann builds this epistemic dependence into a covert operator, the present account locates it in a presupposition, leaving the at-issue

[2] How causal sufficiency is modeled remains a matter of debate, and we do not take a stand here; which framework best suits the present proposal is left for future work.

[3] Koev [18] proposes a similar two-tier, non-modal semantics for Bulgarian evidential statements—a plain at-issue proposition the speaker is committed to, plus a secondary, projecting epistemic layer.

content the bare proposition $\phi$. Indeed, the causation-based presupposition in (4) may well be a special case of a more general felicity condition on assertion: briefly, a proposition $\phi$ is assertable only if the speaker is committed to its truth, and what grounds that commitment varies with the temporal location of $\phi$. If $\phi$ concerns the past or present, the commitment may be grounded in the speaker's observation or experience; if $\phi$ concerns the future, it is grounded in the speaker's present knowledge being causally sufficient for $\phi$.

Additionally, a schedule, a natural law, or a habit provides a well-established causal structure with a chain that is short and reliable: if an event is scheduled, it takes place at the scheduled time; if a natural law holds, its regular consequence follows; if an agent has a standing habit, the habitual action recurs. Because these chains are stable and familiar, a speaker can readily know them in full, and so can readily be in a position of causal sufficiency with respect to their outcomes. It is no accident, then, that these are exactly the cases that license plain futurates.

By contrast, when the current situation fails to be a causally sufficient condition for a proposition's truth, an overt modal becomes necessary, as in (14), repeated here as (43).

| | | | | | |
|---|---|---|---|---|---|
| (43) | Yǒngshì | xiàzhōu | #(huì) | jíbài | Húrén. |
| | the.Warriors | next.week | will | defeat | the.Lakers |
| | 'The Warriors #(will) defeat the Lakers next week.' | | | | |

An unplanned or unscheduled event such as the defeating event in (43) is not causally entailed by the speaker's current knowledge, so its truth value is left undetermined. Specifically, a defeating event is not the terminus of a well-established causal chain. Whether the Warriors defeats the Lakers depends on too many factors (e.g., form, tactics, the opponent's play, chance, and so on) for any of them to constitute a stable structure the speaker could know in advance. Here the speaker's knowledge cannot be causally sufficient for the outcome, because there is no complete chain for the speaker to know.

This is why such events resist the plain futurate and require a modal instead: what a future modal contributes is exactly a way of guaranteeing the outcome that does **not** rest on the speaker's knowing a complete causal chain, i.e., quantifying over inertial worlds in which events are not interrupted and proceed as expected [7, 19, 28].

Finally, the present proposal makes a prediction about negative plain futurates. On the analysis in (42), negation targets only the at-issue proposition, while the causal-sufficiency condition is contributed presuppositionally and therefore projects past it. A negative futurate such as (37)–(38) should thus assert that the event does not occur while still presupposing that the speaker's present knowledge is causally sufficient to settle this—so that the non-occurrence, like the occurrence in a positive futurate, is licensed by inference from what the speaker knows rather than by observation of a future fact. This consequence is welcome, but a full investigation is left for future work.

## 6 Conclusion and Outlook

This paper argues that plain futurates are neither covertly modal nor mere descriptions of a presupposed schedule. Furthermore, it points out a potential direction of understanding plain futurates by placing them alongside dispositional statements, which are assertable on a lawlike or kind-level regularity rather than observation of the instance. Briefly, a plain futurate asserts a simple proposition and presupposes that the speaker's knowledge at speech time is causally sufficient for it. Schedules, natural laws, and habits license plain futurates because each is a short, reliable causal structure that a speaker can know in full; where no such structure exists, causal sufficiency is unavailable and an overt modal must step in to guarantee the outcome without it. This account locates the modal "flavor" of futurates in a presupposition about the speaker's causal knowledge, not in a hidden modal—dissolving the future-contingents puzzle of Section 1 without positing covert modality in the at-issue content. Much is still left for future work. A full causal semantics of plain futurates is reserved for a longer treatment. It should include a precise specification of the causal model and the causal-entailment relation.

**Acknowledgments.** I would like to thank Paul Portner, Rui Guo, Yuyin He, Zhuosi Luo, Xiaolei Fan, Stephen Levinson, Jessica Rett, and Shumian Ye for their discussions and suggestions.

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